

Realtime traffic simulation



USER MANUAL

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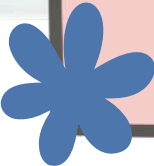


About Our System



A real-time traffic simulation control tool built on top of the SUMO (Simulation of Urban MObility) traffic simulator allows users to observe, control, and experiment with urban traffic scenarios in real time through an interactive graphical interface (GUI).

The system provides an interactive map view of the road network, vehicles, and traffic lights, together with control panels for simulation management, vehicle injection, and traffic light manipulation.



This system is a part of our project in the course of object-oriented programming Java ,and it is designed for learning and experimentation with traffic behavior and control strategies in a simulated urban environment.

Key features and tools

JAVA - Core programming language

TraCI / TraaS - Communication Interface between Java application and SUMO

SUMO (Simulation of Urban MObility)
- Traffic simulation engine

Java Swing - Graphical user interface framework

Key features and tools

KEY FEATURES

- Live connection to a running SUMO simulation
- Real-time map visualization of the road network
- Dynamic display of vehicles and traffic lights
- Zooming and panning within the map view
- Vehicle injection on selected edges and routes
- Batch vehicle injection for testing purposes
- Manual traffic light phase control
- Adjustable traffic light phase durations

Get Started !

important

- The application requires a compatible Java Development Kit (JDK) to run. So that you have to make sure Java is installed correctly and accessible.
- SUMO must be installed on the system. The SUMO executable should be available in the system path so that it can be started directly from the application.
- A valid SUMO configuration file (.sumocfg) is required since it defines the road network, routes, traffic lights, and simulation settings used by the application.



Get Started !

To start the system, follow these steps:

1. Launch the Java application : Run the project from the IDE or execute the compiled JAR file.
2. Choose SUMO execution mode When prompted, select whether SUMO should start:
 - with its native graphical interface (for debugging or comparison)
 - or in background mode (recommended for performance).
3. Establish connection to SUMO : The application automatically connects to the running SUMO instance using the TraCI interface. A successful connection is confirmed in the console output.
4. Start of simulation Once connected, the main control window opens and the simulation begins automatically. Vehicles, traffic lights, and the road network are immediately visible in the map view.

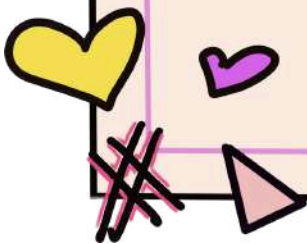
Technical Operation

- **Simulation Control**

- + The simulation can be paused, resumed, or terminated using the control buttons.
- + Simulation speed can be adjusted using the delay slider.

- **Map Interaction**

- + The road network, vehicles, and traffic lights are rendered in real time.
- + Users can zoom and pan the map using the mouse.



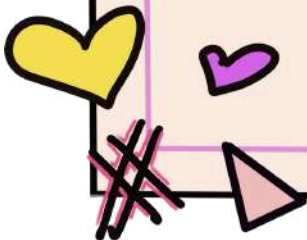
Technical Operation

- **Vehicle Injection**

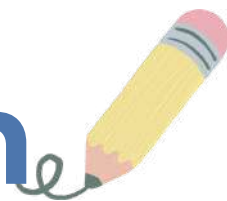
- + Users select a start edge and an available route.
- + Vehicle count, speed, and color can be configured.
- + Injected vehicles appear immediately in the simulation.

- **Traffic Light Management**

- + Traffic lights are displayed with their current states and phases.
- + Users can manually switch phases for a selected traffic light.
- + Phase durations can be adjusted dynamically to observe their effect on traffic flow.



Meet the Team



Each team member contributed to system design, implementation, testing, and documentation.



Duc Thien



Marko



Minh Phu



Hoang Bach



Conclusion

A series of thin, yellow, wavy lines that flow from the left side of the page towards the right, creating a sense of movement and rhythm.

“Having good
time and start
your journey !”

To everyone who played a part in this journey, no
words can adequately express my appreciation !

(author's word)